

# Confusion Matrix — LSTM-AE (Th = 0.015817, FPR(Val) ≤ 5%)

## Rule-Only

| Actual  | Predicted |         |
|---------|-----------|---------|
|         | Normal    | Anomaly |
| Normal  | 5,674     | 49      |
| Anomaly | 318       | 1,918   |

Recall 85.78%   F1 91.27%  
FPR(Attack) 0.86%   FPR(Val) 2.93%

## ML-Only

| Actual  | Predicted |         |
|---------|-----------|---------|
|         | Normal    | Anomaly |
| Normal  | 5,593     | 130     |
| Anomaly | 65        | 2,171   |

Recall 97.09%   F1 95.70%  
FPR(Attack) 2.27%   FPR(Val) 2.71%

## Hybrid

| Actual  | Predicted |         |
|---------|-----------|---------|
|         | Normal    | Anomaly |
| Normal  | 5,585     | 138     |
| Anomaly | 35        | 2,201   |

Recall 98.43%   F1 96.22%  
FPR(Attack) 2.41%   FPR(Val) 4.97%